



Data mining

Customer Segmentation
using RFM Analysist

Team Members:

Ali mostafa

Abdelrahman Hadidi

Katreen Maher

Mahmoud Maher

Mahmoud Ahmed

Nada Ahmed

Presented to:
Dr/Iman Mostafa
Eng/Samaa

Introduction

Customer segmentation is a key technique used in **data analysis** to understand customer behavior and **improve marketing strategies**.

This project applies **RFM** (Recency, Frequency, Monetary) analysis to segment customers based on their purchasing patterns.

Dataset Description

The dataset used is the Online Retail Dataset, which contains transactional data for customers.

Main features include:

- CustomerID
- InvoiceDate
- Quantity
- UnitPrice
- InvoiceNo

This dataset helps analyze how often customers purchase, how recently, and how much they spend.

Data Preprocessing

Before analysis, the dataset was cleaned to ensure accuracy:

- Removed missing values in CustomerID
- Filtered out transactions with negative or zero Quantity
- Removed entries with invalid UnitPrice
- Converted InvoiceDate to datetime format

Feature Engineering

Feature Engineering Feature engineering is the process of creating new meaningful features from raw data to improve model performance and extract deeper insights.

4.1 Total Price per Transaction.

The primary feature created was:

$\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$

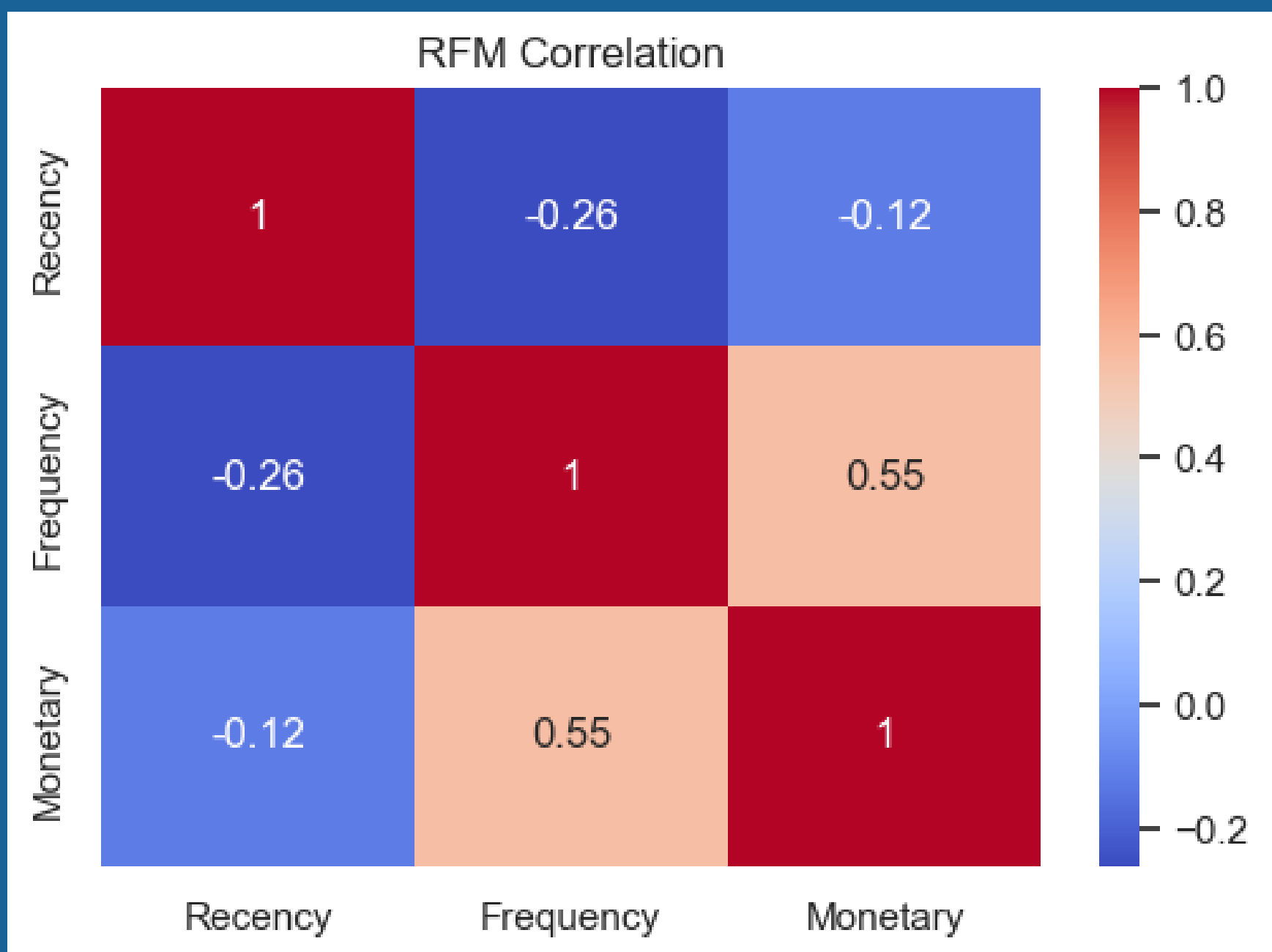
This represents the total spending per transaction and is essential for calculating the Monetary (**M**) component in **RFM** analysis.

RFM Analysis

RFM analysis is based on three metrics:

- **Recency (R)**: Number of days since the last purchase
- **Frequency (F)**: Number of transactions
- **Monetary (M)**: Total amount spent

Customers were grouped by CustomerID and these metrics were calculated for each customer.



Data Scaling

Why Scaling is Necessary:

RFM features have different ranges and units.

Without scaling, large-value features like Monetary dominate K-Means distance calculations, while smaller features like Recency have little impact.

Scaling transforms all features to a similar range so that:

- All RFM features contribute fairly
- No feature dominates the clustering
- Customer segmentation becomes more accurate and balanced

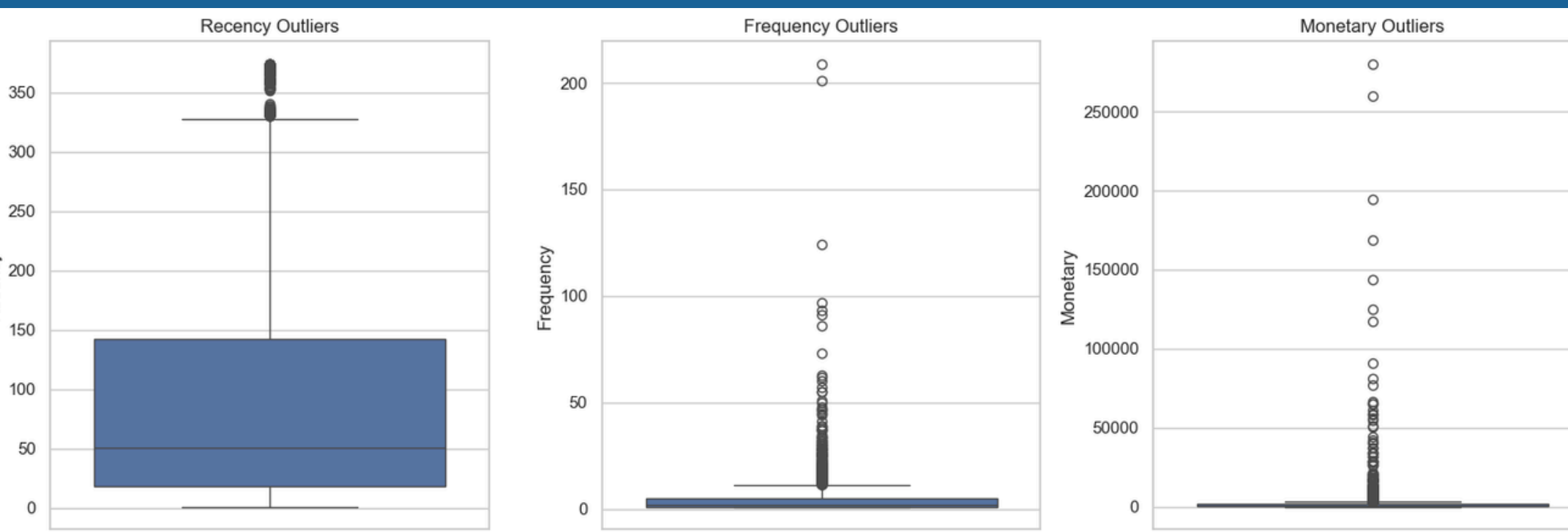
Exploratory Data Analysis (EDA)

Recency Outliers Analysis

- High Recency outliers (150–200+ days) indicate inactive or churned customers.
- Only a small number of customers show extremely high Recency values.
- Recency contains more visible outliers than Frequency and Monetary.

Impact:

Outliers were handled using removal or log transformation to improve clustering quality and K-Means performance.



Log Transformation

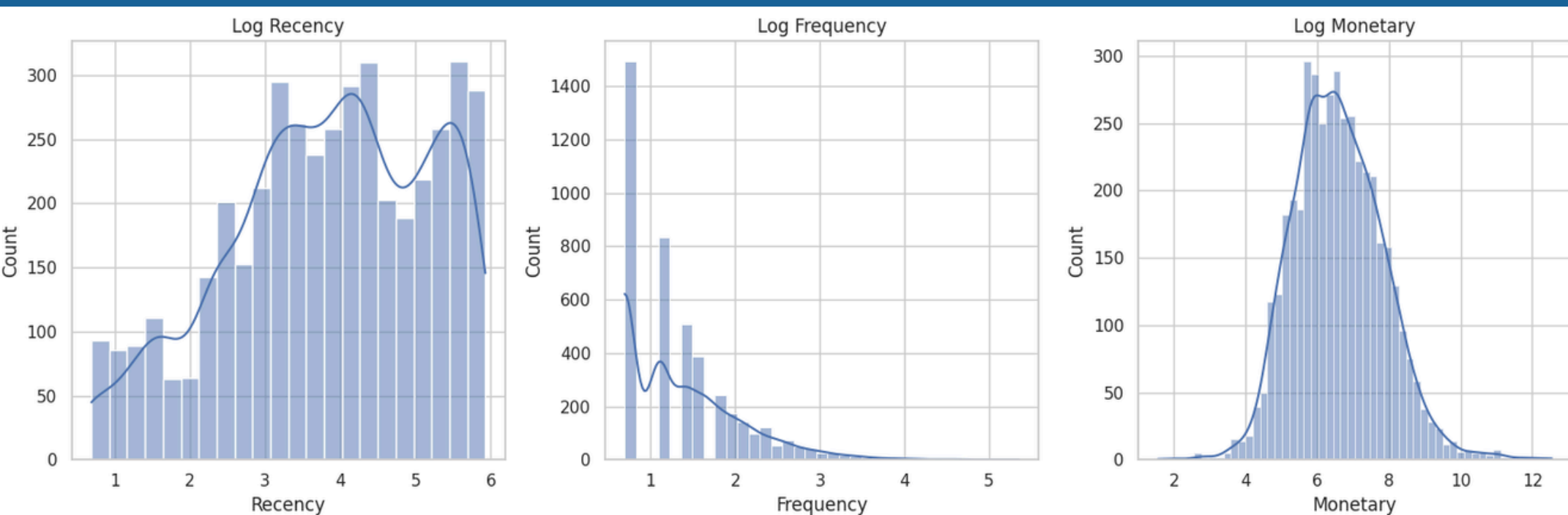
Outlier Detection

the boxplot reveals the presence of high Recency outliers (150–200+ days since the last purchase), representing inactive or churned customers.

Compared to Frequency and Monetary, Recency contains more noticeable extreme values.

Impact:

Outliers were handled using removal or log transformation to improve clustering quality and overall K-Means performance.



Customer Segmentation using RFM Scores

Customers were segmented using quartiles:

- Each metric assigned a score from 1 to 4
- Final RFM Score = $R + F + M$

Example:

- Score 444 → Best customers (loyal & high spenders)
- Score 111 → Low-value customers



K-Means Clustering Results

(Advanced Segmentation)

Cluster	Recency (days)	Frequency	Monetary (Spending)	Segment Name
Cluster 0	41 days	4.8 purchases	2,012	Regular Customers
Cluster 1	246 days	1.5 purchases	631	Churned / Lost Customers
Cluster 2	7 days	80 purchases	122,888	VIP Customers

Business Recommendations

Cluster	Recommendation
Regular Customers (0)	Offer promotions to increase frequency, upsell products
Churned Customers (1)	Send reactivation campaigns, offer discounts to bring them back
VIP Customers (2)	Provide loyalty programs, give exclusive offers, focus on retention (very important group)

Final Results & Insights

- High RFM customers are loyal and profitable
- Low RFM customers are at risk of churn
- Frequency and Monetary strongly impact customer value
- A very small group of customers (VIP) generates the highest revenue
- A large portion of customers are inactive or low-value (Churned segment)
- Most customers fall into the average segment (Regular)
- The business relies heavily on a small number of high-value customers

Segmentation allows targeted marketing campaigns and better resource allocation.

